$$f(x) = \sin\left(2 - \cos\frac{1}{x}\right)$$

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$$\Rightarrow f'(x) = \cos\left(2 - \cos\frac{1}{x}\right) \cdot \left(\sin\frac{1}{x}\right) \cdot \left(\frac{1}{x^2}\right)$$

Simplify:

$$f'(x) = \frac{\left(\cos\left(2 - \cos\left(\frac{1}{x}\right)\right)\right)\left(\sin\left(\frac{1}{x}\right)\right)}{x^2}$$

:D E-Vex